

DSAA Lab 06

Recursion

Areeba (TCT-004)

Thar Institute of Engineering Sciences and Technology (TIEST)

CT-159: Data Structure Algorithms and Applications (Pr)

Engr. Aneel Kumar

October 30, 2025

Exercise

Q1: Print numbers from 5 to 1

Answer:

```
public class Print5to1 {
    static void print(int n){
        if(n==0) return;
        System.out.print(n + " ");
        print(n-1);
    }
    public static void main(String[] args){
        print(5);
    }
}
```

Q2: Print numbers from 1 to 5.

Answer:

```
public class Print1to5 {
    static void print(int n){
        if(n==0) return;
        print(n-1);
        System.out.print(n + " ");
    }
    public static void main(String[] args){
        print(5);
    }
}
```

Q3: Print the sum of first n natural numbers.

Answer:

```
public class SumNatural {
    static int sum(int n){
        if(n==0) return 0;
        return n + sum(n-1);
    }
    public static void main(String[] args){
        System.out.println(sum(5));
    }
}
```

Q4: Print factorial of a number n.

Answer:

```
public class Factorial {
    static int fact(int n){
        if(n<=1) return 1;
        return n * fact(n-1);
    }
    public static void main(String[] args){
        System.out.println(fact(5));
    }
}
```

Q5: Print the fibonacci sequence till nth term.

Answer:

```
public class Fibonacci {
    static int fib(int n){
        if(n<=1) return n;
        return fib(n-1) + fib(n-2);
    }
    public static void main(String[] args){
        int n = 5;
        for(int i=0;i<n;i++)
            System.out.print(fib(i) + " ");
    }
}
```